To Capture, Save, And Analyse Network Traffic on TCP/UDP/IP/HTTP/ARP/DHCP/ICMP/DNS using Wireshark Tool

Aim:

To capture, save, and analyse network traffic using Wireshark for TCP, UDP, IP, HTTP, ARP, DHCP, ICMP, and DNS protocols.

Requirements:

Computer with internet or network connection
Wireshark software
Active network interface

Procedure:

1. Install and open Wireshark
2. Select the active network interface, such as Wi-fi or Ethernet.
3. Click start capturing packets.
4. Generate network traffic by browsing Website, using ping, or performing DNS lookups
5. Apply display filters to examine
6. Stop the capture and save the file
7. Analyse packet headers, source and destination addresses, ports, and protocol details.

Display filters

Protocol	Filter
TCP	tcp
UDP	udp
IP	ip
HTTP	http
ARP	arp
DHCP	dhcp
ICMP	icmp
DNS	dns.

Commands to Generate Traffic

Run these commands in the terminal or Command prompt while wireshark is capturing.

ping 8.8.8.8

nslookup example.com

Result:

Network traffic was captured, saved, and analyzed successfully using Wireshark with protocol-specific display filters for TCP, UDP, IP HTTP, ARP, DHCP, ICMP, and DNS